

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Summer Examination – 2023 Course: B. Tech. Branch : Electronics and Computer/Electronics and Computer Semester : III Science Engineering Subject Code & Name: BTECPC302 Max Marks: 60 Date: 10/08/2023 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Compare Common Source, Common Drain and Common Gate configurations of FET.	4/CO1	6
B)	Explain the construction and working of N-channel E-MOSFET.	2/CO1	6
C)	Discuss the different parameters of JFET.	4/CO1	6
Q.2	Solve Any Two of the following.		12
A)	Explain the block diagram of OP-AMP.	2/CO2	6
B)	Explain with neat diagram Inverting amplifier using OP-AMP.	2/CO2	6
C)	Derive the expression of output voltage for differentiator using OP-AMP.	3/CO2	6
Q. 3	Solve Any Two of the following.		12
A)	Explain the effect of feedback on amplifier characteristics.	2/CO3	6
B)	Illustrate the concept of Cascade amplifier in detail.	3/CO3	6
C)	Draw neat diagram of Hartley Oscillator and explain it in detail.	3/CO3	6
Q.4	Solve Any Two of the following.		12
A)	Describe the three terminal voltage regulators.	2/CO4	6
B)	Explain the Functional block diagram of IC723.	2/CO4	6
C)	Design the voltage regulator for the output voltage of +15 volts and $I_L = 120\text{mA}$ using IC723	6/CO4	6
Q. 5	Solve Any Two of the following.		12
A)	List the different criteria to choose sensors.	2/CO5	6
B)	Explain any one transducer used in measurement of temperature.	2/CO5	6
C)	Explain in detail flow transducer.	2/CO5	6
*** End ***			